

PARANG, Romania

WMO: 153200

Lat: 45.3875N

Lon: 23.4631E

Elev: 5083

StdP: 12.19

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	8.0	-5.7	4.9	19.7	0.5	6.8	15.6	22.1	21.1	18.1	23.8	2.2	0	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	71.5	60.0	69.0	58.8	66.7	58.0	62.3	68.4	60.7	66.8	59.3	65.1	6.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	94.2	65.1	58.4	88.3	63.6	57.0	83.7	62.4	30.8	68.8	29.5	66.9	28.4	65.3	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	13.9	11.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	23.4	26.7	30.1	38.6	47.7	55.0	56.9	58.3	49.3	41.8	35.9	27.2		
	DBStd	14.32	8.98	8.16	8.77	9.71	6.32	6.17	5.69	5.84	8.16	8.90	8.26	8.21		
	HDD50	4139	824	653	617	356	117	25	10	7	115	283	424	708		
	HDD65	8788	1289	1073	1081	792	538	303	255	218	474	720	872	1173		
	CDD50	845	0	0	1	14	44	174	226	263	93	28	2	0		
	CDD65	18	0	0	0	0	0	2	5	9	2	0	0	0		
	CDH74	24	0	0	0	0	0	1	7	13	3	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	3.4	3.0	3.6	3.3	3.8	3.2	3.4	3.5	3.2	3.3	3.1	4.0	3.0		
	PrecAvg	31.3	1.6	1.5	1.8	2.6	3.5	4.3	4.3	2.6	2.9	2.3	2.0	2.0		
	PrecMax	43.5	3.3	3.8	3.6	4.8	6.5	8.0	8.7	5.2	5.9	5.5	6.5	4.3		
(15) Precipitation	PrecMin	19.5	0.4	0.2	0.6	0.6	1.2	0.9	1.6	0.6	0.2	0.0	0.0	0.3		
	PrecStd	6.2	0.8	0.8	0.9	1.1	1.5	1.8	1.9	1.2	1.5	1.3	1.4	1.1		
	DB	46.5	48.2	53.5	62.9	65.3	71.7	73.7	75.1	72.7	64.1	57.5	51.5			
	MCWB	34.0	40.6	41.6	48.5	54.7	60.0	60.2	61.0	60.7	49.5	43.1	36.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	40.4	43.5	48.9	58.9	62.3	68.9	70.4	72.0	66.9	60.9	53.0	44.4			
	2% MCWB	33.3	37.5	39.9	48.2	53.3	60.4	59.7	60.4	55.1	49.7	42.5	35.5			
	5% DB	37.3	39.7	45.2	55.2	59.4	66.5	68.0	69.0	63.1	57.1	49.8	40.5			
	5% MCWB	32.9	34.7	37.2	45.6	51.9	59.4	59.5	58.4	54.4	47.7	41.8	35.1			
	10% DB	35.1	37.1	42.0	51.5	56.8	64.2	65.5	66.5	60.3	54.1	46.7	37.2			
	10% MCWB	31.3	33.6	36.1	43.5	50.4	57.7	58.0	57.2	52.7	46.1	40.9	33.3			
	0.4% WB	37.8	42.3	43.6	51.5	57.8	64.5	63.7	64.0	60.3	54.3	48.0	40.5			
	0.4% MCDB	41.2	45.6	51.3	60.0	63.9	68.5	69.5	71.9	69.3	58.5	54.8	44.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.1	38.8	40.7	48.6	54.6	62.1	61.9	61.7	57.4	51.3	45.7	37.5			
	2% MCDB	38.6	42.6	47.0	57.3	60.2	66.9	67.7	68.8	64.4	57.2	50.1	42.5			
	5% WB	34.2	36.1	38.6	46.7	52.7	60.2	60.3	60.1	55.3	49.4	44.2	35.5			
	5% MCDB	36.7	39.1	43.7	53.7	58.3	65.9	66.0	66.3	61.9	55.6	47.7	39.5			
	10% WB	32.2	34.1	36.5	44.6	50.9	58.3	59.0	58.8	53.5	47.7	42.5	33.9			
	10% MCDB	34.4	36.3	40.9	51.4	56.1	63.5	64.5	64.7	58.8	53.1	45.5	36.6			
	MDBR	10.9	10.6	11.1	12.0	11.2	11.8	12.3	11.4	10.6	10.3	9.6	9.5			
	5% DB	14.4	12.6	14.5	15.4	14.4	14.4	14.6	13.2	13.2	13.6	12.3	12.3			
(36) Mean Daily Temperature Range	MCWBR	8.9	8.5	8.7	9.3	9.4	9.8	9.2	8.2	8.2	10.1	7.6	7.6			
	MCDBR	11.1	11.2	13.0	14.8	13.5	14.0	13.1	12.2	12.5	12.1	9.5	10.6			
	MCWBR	8.1	8.2	8.7	9.2	9.2	10.2	9.2	8.6	8.5	9.6	6.5	7.9			
	MCDBR	8.1	8.2	8.7	9.2	9.2	10.2	9.2	8.6	8.5	9.6	6.5	7.9			
(40) Clear-Sky Solar Irradiance	taub	0.262	0.286	0.320	0.368	0.357	0.355	0.368	0.379	0.344	0.314	0.292	0.265			
	taud	2.411	2.358	2.356	2.276	2.351	2.378	2.337	2.304	2.391	2.484	2.498	2.454			
	Ebn at Noon	274	285	289	282	289	289	283	276	277	272	260	261			
	Edh at Noon	23	29	34	40	38	37	38	38	32	25	21	19			
(44) All-Sky Solar Radiation	RadAvg	422	674	1028	1386	1651	1858	1834	1663	1186	825	487	347			
	RadStd	54	87	141	153	149	175	150	161	120	97	61	45			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	+1.15	N/A	N/A	N/A	+1.60	N/A	N/A	-205	-302	+164	+79

Nomenclature: See separate page